

Project Plan

Modular Power Electronics & Current Control for the Geec

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Revision	1

1 Scope Statement

The scope of this project will include:

1. Evaluation of Current Power Electronics on the Geec
2. Evaluation of Brushless versus Brushed Motor
3. Simulation & Design of Power Electronics Board
4. Design of Power Control Board
5. Verification of safe functionality & implementation on the Geec

2 Stakeholders

Stakeholder	Power	Interest	Concerns
Supervisor (Owner)	High	High	On time & achieve milestones/outputs Timely Solutions to project issues Regular Reports, safe and reliable implementation of deliverables
Co-Assessor	Medium	Low	Achieve Deliverables
Student(me)	High	High	Technical feasibility, meeting project scope and deadlines, producing results
Geec Team	Medium	High	
SEM Organiser	High	High	Design is compliant with the rules as written by SEMD
Technical Officers	Medium	Medium	Safe use of lab equipment
EEE Department	High	Medium	Compliance with academic standards, successful project outcome
Geec Driver	Medium	High	System needs to be responsive and fail safe for driver safety

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3 Deliverables

3.1 Fixed Deliverables

- Project Plan
- Progress Report & Presentation
- Project Dissemination (Word Document + PPT document)
- Bench Demonstration
- Final Project Report

3.2 Interim Deliverables

- Hardware Simulation
- Control Simulation
- Power Electronics Schematic and Layout
- Power Controller Schematic and Layout

4 Work Breakdown Structure

4.1 WP1: Background Research and Evaluation of Existing System

Deliverable: This Work Package should allow for an informed decision on the benefits and drawbacks of the current power electronics and the required direction forward for the Geec.

4.1.1 Task 1: BLDC Evaluation

- Rory Egan's thesis last year investigated the use of a BLDC motor, suggesting that there is much greater performance to be achieved using BLDC.
- Investigate high efficiency BLDC motors in the 100-1000W range.
- Compare using existing data and manufacturer specifications the energy requirements.

4.1.2 Investigate the Current Power Electronics Design

- Evaluate the parts and layout of the current power electronics.
- Identify the capabilities and limitations involved in the current design.
- Simulate the current design.

4.1.3 Research different methods of control, power etc.

- Investigate the newer power MOSFETs, Gate Drivers, HEMTs and sensors.
- Investigate different control strategies, open loop, hardware vs software closed loop.
- Investigate connectors separate solutions.

4.2 WP2: Power Electronics Design

Deliverable: Power Electronics board with gate driver, Power MOSFETs, Current Sensors, Bulk Capacitors and non-isolated

4.2.1 Task 1: Power Stage Circuit Simulation

- Create an accurate simulation of the new power electronics.
- Determine the power consumption during switching events.
- Evaluate current ripple due to different switching frequencies

4.2.2 Task 2: Power Electronics Part Selection

- Select and Simulate other electrical parts. Current sensors, voltage sensors, combinations of capacitors (electrolytic, ceramic, Film, MLCC).
- Create a block diagram and of parts on each board.
- Estimate Power Consumption for idle and active states.

4.2.3 Task 3: Schematic Design

- Create part library in Altium with easy-to-understand symbols
- Create a 1- or 2-page schematic showcasing all components keeping all components that are functionally close to each other.
- Use SMD devices 0805 or greater in size.

4.2.4 Task 4: Layout Design

- Import schematic into PCB designer.
- Design 4-layer board keeping a large ground plane and minimising loop areas in switch nodes. Have separate ground planes for high resolution analogue signals vs power plane
- Order and assemble multiple boards.

4.3 WP3: Control Electronics Design

Deliverable: Power Controller board able to interface directly with the power electronics board. Containing micro-controller, hardware signal conditioning, DCDC converters, signal isolation.

4.3.1 Task 1: Control Scheme Comparison

Evaluate different control schemes: Open Loop, Closed Loop in hardware, software and hybrid approach.

4.3.2 Task 2: Hardware Control Simulation

- Simulate the chosen control scheme using a simplified power stage
- Simulate the hardware sensors and controllers

4.3.3 Task 3: Control Electronics Schematic

- Design a 1- or 2-page schematic with all components divided into functional block areas.
- Integrate isolation, DCDC conversion and signal conditioning.
- Use standard and reliable connectors to the power electronics board. Integrate extra connectors to integrate a DAQ board or other communication protocols

4.3.4 Task 4: Control Electronics Layout

- Import schematic into PCB designer
- Design 2-layer board creating a large and continuous ground plane where possible.
- Add mounting holes and order PCBs to be assembled in Galway.

4.4 WP4: Electrical Verification

Deliverable: Verify the functionality of both the power electronics board and the power controller board.

4.4.1 Open Loop Power Electronic Testing

- Verify bridge functionality behaves as per simulations
- The Power Electronics board can be tested open loop to verify functionality and characterise the device
- Determine maximum specifications, voltage, PWM % current etc.

4.4.2 Power Controller Testing

- Verify the functionality of the hardware/software controller
- Determine isolation efficacy, response speed for current controller.

4.4.3 Closed Loop Motor Control

- Connect the power electronics board to the power controller
- Run an Arduino script connecting the throttle and running an unloaded motor.
- Use a very low maximum current and use an oscilloscope to see the current limiting.

4.4.4 Verification on Mini Dynamometer

- In combination with Seán Cotter's mini dynamometer project verify functionality with different current limits on different loads
- Identify stall torques and currents and use the current limiting to meet the limits.

4.5 WP5: Implementation on the Geec

Deliverable: Power Electronics system integrated into the Geec capable of racing at SEM

4.5.1 Integrate Power Electronics into the Geec

- Create all necessary connections for the new power electronics board
- Add connections to the steering wheel and DAQ.

4.5.2 Run new Power Electronic on the Dynamometer

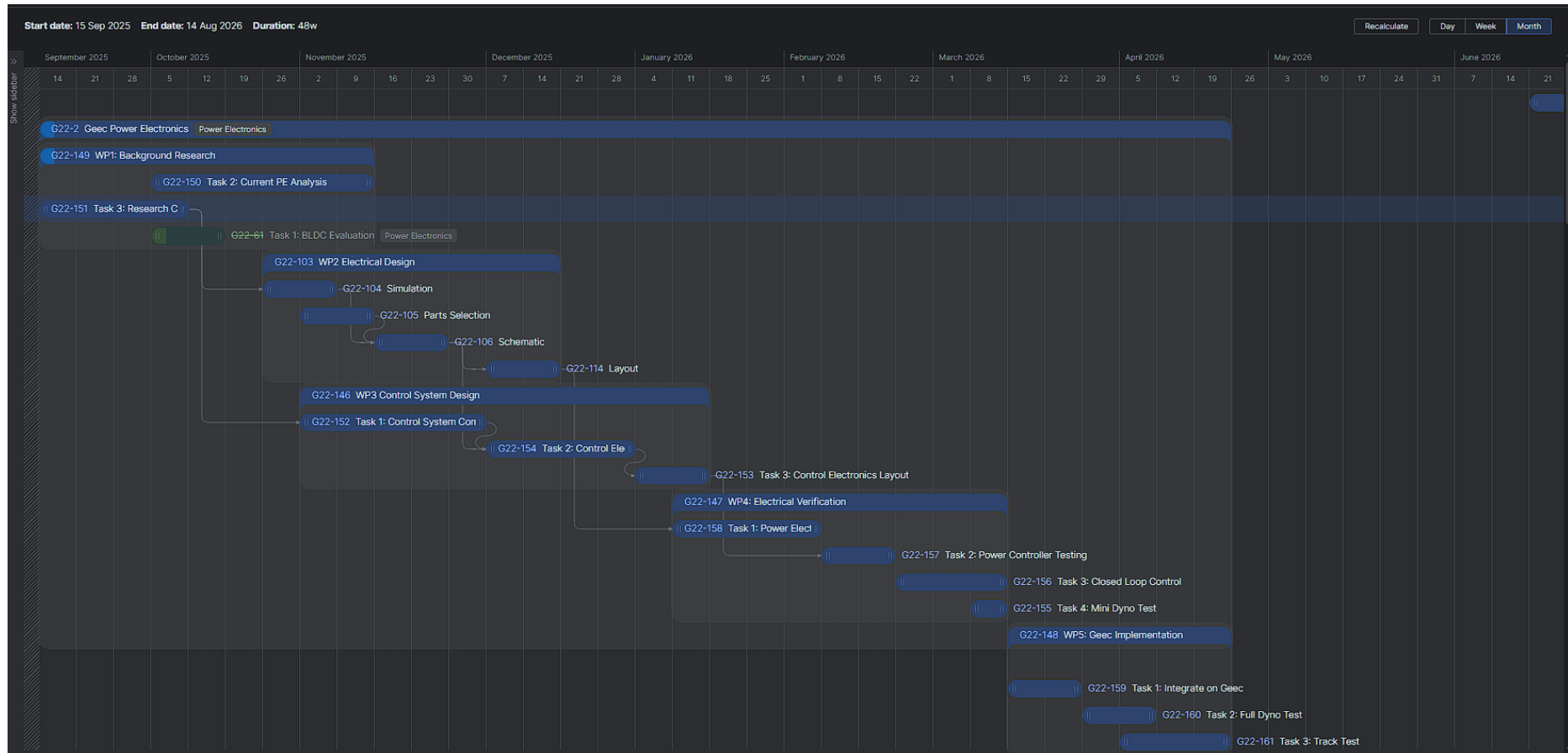
- Complete a safe test of the new power electronics on the Dynamometer and compare the results with existing dynamometer data
- Run a variety of torques and current limits

4.5.3 Complete a Track Test

- Complete a track test in Dangan or another location and run a full race as per SEM rules.
- Complete a simulated technical inspection of the electrical system based on

5 Work Schedule

The work schedule for the project is being managed alongside the other team members in YouTrack, details of completed work will be kept in tasks and subtasks at <https://thegeec.youtrack.cloud/gantt-charts/210-1> Below is a screenshot of the Gantt Chart for the above project



6 Risk Register

The project will be coordinated with respect to existing Geec SOPs and RAs, necessary SOPs and RAs will be updated as required:

Title	Severity	Likelihood	Overall Risk
Battery Charging			
Power Electronics Testing	Low	Low	Low
Testing On Dynamometer	Medium	Low	Medium
Testing on Mini-Dynamometer	Medium	Low	Medium
Track Testing	Medium	Low	Medium
Manual Tools	Low	Low	Low
Power Tools	Medium	Low	Medium

All Risk assessments are available in the Geec lab (ENG-G011) and on the Geec SharePoint.

7 Communication Strategy

Name	Recipient	Frequency	Medium	Contents
Supervisor Meetings	Supervisor	Weekly (Monday)	In Person	Discuss actions of the previous week and next week
Leads Meeting	Teams Leads and Lecturers	Weekly (Tuesday)	In Person	Major Events on the team
YouTrack Tasks	Supervisor and Team	As work is Completed	YouTrack	Technical Details on work done
Progress Report	Supervisor and Co-Assessor	Once (deliverable)	In Person	PowerPoint and Report
Poster	All Stakeholder	Once (deliverable)	Public Event	Project Achievements
Bench Demo	Supervisor and Co-Assessor	Once (deliverable)	In Person	Demonstration of Technical Achievements
Final Report	Supervisor and Co-Assessor	Once (deliverable)	Document	In depth details on Technical Work